Lemma: G是群, $A \leq G$, $B \leq G$, 若 $A \subseteq B$, 则 $A \leq B$.

Proof: $\because B \leq G$ $\therefore (B, \cdot)$是群, $1_B = 1_G$.

$\because A \leq G$ $\therefore 1_G = 1_A$ $\therefore 1_A = 1_G = 1_B$.

$\therefore (B, \cdot)$是群, $A \subseteq B$, $1_B = 1_A \in A$

$\because A \leq G$ $\therefore A$对乘法封闭, A对取逆封闭.

$\therefore A \leq B$. $\square$

定义（群的子集的乘积）G是群，对$\forall A, B \subseteq G$，引入记号：

$$AB = \{ab \in G : a \in A \text{ 且 } b \in B\} \subseteq G$$

Lemma：G是群，$H \leqslant G$，$K \leqslant G$，则有：

$$HK = KH \iff HK \leqslant G$$

Proof：$(\Rightarrow)$：$\because H \subseteq G, K \subseteq G \quad \therefore HK \subseteq G$.

$\because H \leqslant G$ 且 $K \leqslant G \quad \therefore 1_G \in H$ 且 $1_G \in K$

$\therefore 1_G = 1_G \cdot 1_G \in HK$

任取HK中的两个元素：h_1k_1，h_2k_2（其中$h_1, h_2 \in H$，$k_1, k_2 \in K$.

$\because k_1h_2 \in KH = HK \quad \therefore \exists h_3 \in H, k_3 \in K$，s.t. $k_1h_2 = h_3k_3$

$\therefore (h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2 = h_1(h_3k_3)k_2 = (h_1h_3)(k_3k_2) \in HK$

$\therefore HK$对乘法封闭.

对$\forall hk \in HK$（其中$h \in H, k \in K$）

$(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$

$\therefore HK$对取逆封闭 $\quad \therefore HK \leqslant G$.

$(\Leftarrow)$：$\because H \subseteq G$ 且 $K \subseteq G \quad \therefore HK \subseteq G$

$\because K \subseteq G$ 且 $H \subseteq G \quad \therefore KH \subseteq G$

对$\forall x \in HK$，$\because HK \leqslant G \quad \therefore x^{-1} \in HK$

$\therefore \exists h\in H,\ k\in K,\ s.t.\ x^{-1}=hk$

$\therefore x=(x^{-1})^{-1}=(hk)^{-1}=k^{-1}h^{-1}\in KH$ $\quad\therefore HK\subseteq KH$.

对$\forall x\in KH$, $\exists k\in K,\ h\in H,\ s.t.\ x=kh$

$\therefore x^{-1}=(kh)^{-1}=h^{-1}k^{-1}\in HK$ $\quad\because HK\leqslant G$ $\quad\therefore (x^{-1})^{-1}\in HK$

$\therefore x\in HK$ $\quad\therefore KH\subseteq HK\subseteq KH$ $\quad\therefore HK=KH$ $\quad\square$

Lemma: G是群, $H\leqslant G$, $K\lhd G$, 则有: $HK=KH$且$HK\leqslant G$

~~证~~ proof: 对$\forall hk\in HK$ (其中$h\in H$, $k\in K$), 有:

$\because h\in G,\ k\in K,\ K\lhd G$ $\quad\therefore hkh^{-1}\in K$

$\therefore hk=(hk)(h^{-1}h)=(hkh^{-1})h\in KH$ $\quad\therefore HK\subseteq KH$

对$\forall kh\in KH$ (其中$k\in K$且$h\in H$), 有:

$\because h\in H$ $\quad\therefore h\in G$ $\quad\therefore h^{-1}\in G$.

$\because h^{-1}\in G,\ k\in K,\ K\lhd G$ $\quad\therefore h^{-1}k(h^{-1})^{-1}\in K$ $\quad\therefore h^{-1}kh\in K$

$\therefore kh=(hh^{-1})(kh)=h(h^{-1}kh)\in HK$ $\quad\therefore KH\subseteq HK$

$\therefore HK=KH$ $\quad\therefore HK\leqslant G$. $\quad\square$

定义(中心化子, 正规化子) G是群, $E\subseteq G$是G的任一子集,

定义E在G中的中心化子为$Z_G(E):=\{\alpha\in G:$ 对$\forall x\in E$, 都有$x\alpha=\alpha x\}$

定义E在G中的正规化子为$N_G(E):=\{\alpha\in G: \alpha E\alpha^{-1}=E\}$

当$E=\{x\}$时, 记$Z_G(E)=Z_G(x)$, $N_G(E)=N_G(x)$

Lemma: G是群，$E \subseteq G$，则有：

① $Z_G(E) \leqslant G$

② $N_G(E) \leqslant G$

③ $Z_G(E) \lhd N_G(E)$

Proof: $\because 1_G \in G$，且对$\forall x \in E$，有：$x \cdot 1_G = x = 1_G \cdot x$ ~~$\therefore 1_G \in Z_G(E)$~~

$\therefore 1_G \in Z_G(E)$，显然 $Z_G(E) \subseteq G$

对$\forall \alpha, \beta \in Z_G(E)$，有：$\alpha, \beta \in G$ $\quad \therefore \alpha\beta \in G$.

对$\forall x \in E$，有：$x(\alpha\beta) = (x\alpha)\beta = (\alpha x)\beta = \alpha(x\beta)$

$$= \alpha(\beta x) = (\alpha\beta)x$$

$\therefore \alpha\beta \in Z_G(E)$ $\quad \therefore Z_G(E)$对乘法封闭.

对$\forall \alpha \in Z_G(E)$，有：$\alpha \in G$ $\quad \therefore \alpha^{-1} \in G$.

对$\forall x \in E$，有：$x\alpha^{-1} = (\alpha^{-1}\alpha)(x\alpha^{-1}) = \alpha^{-1}(\alpha x)\alpha^{-1}$

$$= \alpha^{-1}(x\alpha)\alpha^{-1} = (\alpha^{-1}x)(\alpha\alpha^{-1}) = \alpha^{-1}x$$

$\therefore \alpha^{-1} \in Z_G(E)$ $\quad \therefore Z_G(E)$对取逆封闭

$\therefore Z_G(E) \leqslant G$. ①得证.

$\because 1_G \in G$ 且 $1_G E 1_G^{-1} = 1_G E 1_G = \{1_G x 1_G : x \in E\} = \{x : x \in E\}$

$$= E.$$

$\therefore 1_G \in N_G(E)$. 显然 $N_G(E) \subseteq G$.

对$\forall \alpha, \beta \in N_G(E)$，有：$\alpha, \beta \in G$ $\therefore \alpha\beta \in G$.

任取$(\alpha\beta)E(\alpha\beta)^{-1}$中的一元：$(\alpha\beta)x(\alpha\beta)^{-1}$ （其中$x \in E$）

$\because (\alpha\beta)x(\alpha\beta)^{-1} = \alpha\beta x \beta^{-1}\alpha^{-1} \in E$. $\therefore (\alpha\beta)E(\alpha\beta)^{-1} \subseteq E$.

对$\forall x \in E$，$\because \alpha E \alpha^{-1} = E$ $\therefore \exists e_1 \in E$, s.t. $x = \alpha e_1 \alpha^{-1}$

$\because \beta E \beta^{-1} = E$ $\therefore \exists e_2 \in E$, s.t. $e_1 = \beta e_2 \beta^{-1}$

$\therefore x = \alpha e_1 \alpha^{-1} = \alpha(\beta e_2 \beta^{-1})\alpha^{-1} = (\alpha\beta)e_2(\alpha\beta)^{-1} \in (\alpha\beta)E(\alpha\beta)^{-1}$

$\therefore E \subseteq (\alpha\beta)E(\alpha\beta)^{-1} \subseteq E$ $\therefore (\alpha\beta)E(\alpha\beta)^{-1} = E$

$\therefore \alpha\beta \in N_G(E)$ $\therefore N_G(E)$对乘法封闭.

对$\forall \alpha \in N_G(E)$，有：$\alpha \in G$ $\therefore \alpha^{-1} \in G$.

任取$\alpha^{-1}E\alpha$中的一元：$\alpha^{-1}x\alpha$ （其中$x \in E$）.

$\because \alpha E \alpha^{-1} = E$ $\therefore \exists e_1 \in E$, s.t. $x = \alpha e_1 \alpha^{-1}$

$\therefore \alpha^{-1}x\alpha = \alpha^{-1}(\alpha e_1 \alpha^{-1})\alpha = (\alpha^{-1}\alpha)e_1(\alpha^{-1}\alpha) = e_1 \in E$.

$\therefore \alpha^{-1}E\alpha \subseteq E$.

对$\forall x \in E$，$\because \alpha x \alpha^{-1} \in \alpha E \alpha^{-1} = E$.

$\therefore \alpha^{-1}(\alpha x \alpha^{-1})\alpha \in \alpha^{-1}E\alpha$

$\because \alpha^{-1}(\alpha x \alpha^{-1})\alpha = (\alpha^{-1}\alpha)x(\alpha^{-1}\alpha) = 1_G x 1_G = x$

$\therefore x \in \alpha^{-1}E\alpha$ $\therefore E \subseteq \alpha^{-1}E\alpha \subseteq E$ $\therefore \alpha^{-1}E\alpha = E$

$\therefore \alpha^{-1}E(\alpha^{-1})^{-1} = E$ $\therefore \alpha^{-1} \in N_G(E)$ $\therefore N_G(E)$对取逆封闭 $\therefore N_G(E) \leqslant G$

②得证.

对$\forall \alpha\in Z_G(E)$，有：$\alpha\in G$.

$\because \alpha E\alpha^{-1}=\{\alpha x\alpha^{-1}: x\in E\}=\{x\alpha\alpha^{-1}: x\in E\}=\{x: x\in E\}$

$=E$

$\therefore \alpha\in N_G(E)$ $\quad\therefore Z_G(E)\subseteq N_G(E)$ $\quad\therefore Z_G(E)\leqslant N_G(E)$.

~~又~~ $\because (N_G(E),\cdot)$是群，$Z_G(E)\leqslant N_G(E)$

对$\forall \alpha\in N_G(E)$，有：任取 $\alpha Z_G(E)\alpha^{-1}$ 中的一元：$\alpha\lambda\alpha^{-1}$

（其中 $\lambda\in Z_G(E)$）

$\because \alpha\in N_G(E)$ $\quad\therefore \alpha\in G$ $\quad\because \lambda\in Z_G(E)$ $\quad\therefore \lambda\in G$ $\quad\therefore \alpha\lambda\alpha^{-1}\in G$

对$\forall x\in E$，$\because \alpha\in N_G(E)$，$N_G(E)\leqslant G$ $\quad\therefore \alpha^{-1}\in N_G(E)$

$\therefore \alpha^{-1}E(\alpha^{-1})^{-1}=E$ $\quad\therefore \alpha^{-1}E\alpha=E$

$\therefore \alpha^{-1}x\alpha\in\alpha^{-1}E\alpha=E$.

$\because \alpha^{-1}x\alpha\in E$，$\lambda\in Z_G(E)$ $\quad\therefore (\alpha^{-1}x\alpha)\lambda=\lambda(\alpha^{-1}x\alpha)$

$\therefore \alpha(\alpha^{-1}x\alpha\lambda)\alpha^{-1}=\alpha(\lambda\alpha^{-1}x\alpha)\alpha^{-1}$

$\therefore x(\alpha\lambda\alpha^{-1})=(\alpha\lambda\alpha^{-1})x$ $\quad\therefore \alpha\lambda\alpha^{-1}\in Z_G(E)$

$\therefore \alpha Z_G(E)\alpha^{-1}\subseteq Z_G(E)$ $\quad\therefore Z_G(E)\lhd N_G(E)$，③得证. $\square$

Lemma：G是群，$K\leqslant G$，则有：$K\lhd N_G(K)$.

proof：$\because K\leqslant G$ $\quad\therefore K\subseteq G$ $\quad\therefore N_G(K)\leqslant G$.

对$\forall \alpha\in K$，有：$\alpha\in G$. $\because K\subseteq G,\ \alpha\in G$ $\quad\therefore \alpha K\alpha^{-1}\subseteq G$.

任取$\alpha K\alpha^{-1}$中的一元：$\alpha k\alpha^{-1}$（其中$k\in K$）.

$\because K\leq G,\ \alpha\in K,\ k\in K \quad \therefore \alpha k\alpha^{-1}\in K \quad \therefore \alpha K\alpha^{-1}\subseteq K$.

对$\forall k\in K$，$\because K\leq G,\ \alpha\in K,\ k\in K \quad \therefore \alpha^{-1}k\alpha\in K$.

$\therefore \alpha(\alpha^{-1}k\alpha)\alpha^{-1}\in \alpha K\alpha^{-1}$

$\because \alpha(\alpha^{-1}k\alpha)\alpha^{-1}=(\alpha\alpha^{-1})k(\alpha\alpha^{-1})=k \quad \therefore k\in\alpha K\alpha^{-1}$

$\therefore K\subseteq \alpha K\alpha^{-1}\subseteq K \quad \therefore \alpha K\alpha^{-1}=K$

$\therefore \alpha\in N_G(K) \quad \therefore K\subseteq N_G(K) \quad \therefore K\leq N_G(K)$.

已知$(N_G(K),\cdot)$是群，$K\leq N_G(K)$

对$\forall \alpha\in N_G(K)$，有：$\alpha\in G$且$\alpha K\alpha^{-1}=K \quad \therefore \alpha K\alpha^{-1}\subseteq K$

$\therefore K\lhd N_G(K)$. $\square$

Lemma：G是群，$K\leq G$，若$L\leq G$且$K\leq L$，则有：

$$K\lhd L \iff L\leq N_G(K)$$

Proof：$(\Rightarrow)$：$\because L\leq G \quad \therefore (L,\cdot)$是群.

$\therefore (L,\cdot)$是群，$K\leq L$. $\cancel{\therefore K\lhd L}$

~~$\therefore$对$\forall\alpha\in L$，都有$\alpha K\alpha^{-1}=K$.~~

对$\forall \alpha\in L$. $\because K\lhd L \quad \therefore \alpha K\alpha^{-1}=K$.

$\therefore \alpha\in G$ 且 $\alpha K\alpha^{-1}=K$ $\therefore \alpha\in N_G(K)$ $\therefore L\subseteq N_G(K)$.

$\therefore L\leq N_G(K)$

$(\Leftarrow)$: $\because L\leq G$ $\therefore (L,\cdot)$是群.

$\therefore (L,\cdot)$是群且 $K\leq L$.

对$\forall \alpha\in L$, $\because L\leq N_G(K)$ $\therefore \alpha\in N_G(K)$.

$\therefore \alpha\in G$ 且 $\alpha K\alpha^{-1}=K$. $\therefore K\lhd L$. $\square$

推论: G是群, $K\leq G$, 则有: $N_G(K)$是G中最大的使K是其正规子群的子群.

Proof: 由上一引理显然. $\square$.

Lemma: G是群, $K\leq G$, 则有:

$$K\lhd G \iff N_G(K)=G.$$

Proof: $\because G$是群, $K\leq G$, $G\leq G$, $K\leq G$.

$\therefore K\lhd G \iff G\leq N_G(K) \iff N_G(K)=G$ $\square$

Lemma: G是群, $H\leq G$, $K\leq G$, $H\subseteq N_G(K)$, 则有: $HK=KH$ 且 $HK\leq G$, 且 $K\lhd HK$.

Proof: 对$\forall x\in HK$, $\exists h\in H$ 且 $k\in K$, 有: $x=hk$.

$\because h\in H\subseteq N_G(K)$ $\therefore h\in N_G(K)$ $\therefore h\in G$ 且 $hKh^{-1}=K$.

$\because k\in K \quad \therefore hkh^{-1}\in hKh^{-1}=K$

$\therefore x=hk=(hk)(h^{-1}h)=(hkh^{-1})h\in KH \qquad \therefore HK\subseteq KH$

对$\forall x\in KH$，$\exists k\in K$，$h\in H$，s.t. $x=kh$.

$\because H\leqslant G$，$h\in H \quad \therefore h^{-1}\in H \quad \because H\subseteq N_G(K) \quad \therefore h^{-1}\in N_G(K)$

$\therefore h^{-1}\in G$且 $h^{-1}K(h^{-1})^{-1}=K \quad \therefore h^{-1}Kh=K$

$\because k\in K \quad \therefore h^{-1}kh\in h^{-1}Kh=K$.

$\therefore x=kh=(hh^{-1})kh=h(h^{-1}kh)\in HK \quad \therefore KH\subseteq HK$

$\therefore HK=KH$. $\quad \therefore HK\leqslant G$.

对$\forall k\in K$，有：$k\in G \quad \therefore k=1_G\cdot k\in HK \quad \therefore K\subseteq HK$

$\therefore K\leqslant HK$

已知$(HK,\cdot)$是群，$K\leqslant HK$.

对$\forall x\in HK$，任取 xKx^{-1}中的一元：xkx^{-1}（其中$k\in K$）

$\because x\in HK \quad \therefore \exists \alpha\in H$，$\beta\in K$，s.t. $x=\alpha\beta$.

$\because \beta\in K$，$k\in K$，$K\leqslant G \quad \therefore \beta k\beta^{-1}\in K$.

$\because \alpha\in H$，$H\subseteq N_G(K) \quad \therefore \alpha\in N_G(K) \quad \therefore \alpha\in G$且$\alpha K\alpha^{-1}=K$

$\therefore xkx^{-1}=(\alpha\beta)k(\alpha\beta)^{-1}=\alpha\beta k\beta^{-1}\alpha^{-1}=\alpha(\beta k\beta^{-1})\alpha^{-1}\in\alpha K\alpha^{-1}=K$

$\therefore xKx^{-1}\subseteq K \quad \therefore K\lhd HK$. $\quad \square$

Lemma：G是群，$H \leqslant G$，$K \lhd G$，则有：$HK = KH$ 且 $HK \leqslant G$ 且 $K \lhd HK$.

Proof：$\because$ G是群，$K \lhd G$ $\quad \therefore N_G(K) = G$

$\because H \leqslant G \quad \therefore H \leqslant N_G(K) \quad \therefore H \subseteq N_G(K)$

$\therefore HK = KH$ 且 $HK \leqslant G$ 且 $K \lhd HK$. $\quad \square$